

SIH ORIGIN

Engineering Execution OS & Architecture Blueprint

Problem Statement ID: 26034 | **Theme:** Miscellaneous / Software
Title: Software System to Check Compliance of Packaged Commodities under Legal Metrology Rules, 2011

EXECUTIVE OPERATIONAL MANDATE

This document serves as the single source of truth and practical operating system for **Team IRIS** to execute, code, integrate, test, and present the **SIH Origin** platform during the 24-Hour Smart India Hackathon. Built with **Google Antigravity** as the primary autonomous agentic engineering engine, this blueprint establishes strict phase gates, prompt sequences, data contracts, and judge-aligned demo pathways.

PROJECT DETAILS	TEAM IRIS SPECIFICATIONS
Project Codename: SIH Origin Event: Smart India Hackathon 2026 PS Category: Software (Government Compliance) Target Domain: Legal Metrology Enforcement Primary Tool: Google Antigravity Agentic IDE Target Build Time: 24 Hours (Continuous Sprint)	Team Leader: Karthik V (RA2411056010201 - DSBS) Team Members: • Shrihan P (DSBS) • Akshith Varma A (DSBS) • Dhanush Reddy C (DSBS) • Dharshini K (DSBS) • Yashwanth Raju (CINTEL) Mentors: Dr. T. Ragupathi & Sravya A

CORE ENGINEERING PRINCIPLES FOR THE HACKATHON

- One Signature End-to-End Flow First:** Never build disconnected features. Build the full loop (Image Ingestion → OCR Extraction → Rule Verification → Evidence Bounding Box Overlay → Inspection Report).
- Antigravity-First Modularity:** Never execute massive monolithic prompts. Execute atomic vertical slices (Schema → Backend Engine → Typed API → UI Components → Integration).
- Zero-Fail Demo Safeguards:** Seed deterministic backup sample data to insulate live presentations from cloud network jitter and external rate limits.

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2. EXECUTIVE SUMMARY & CORE PRODUCT VISION

The Challenge: In India, millions of packaged consumer goods are sold daily across physical retail and e-commerce platforms. The Department of Consumer Affairs mandates strict adherence to the *Legal Metrology (Packaged Commodities) Rules, 2011*. However, compliance enforcement is currently crippled by manual inspections, slow auditing workflows, subjective errors, and inability to parse multilingual, curved, or low-contrast packaging at scale.

The SIH Origin Solution: SIH Origin is an intelligent, automated regulatory compliance screening platform. It ingests package label photographs or digital e-commerce assets, applies adaptive computer vision and multilingual Optical Character Recognition (OCR), extracts all mandatory declarations, validates them against a codified Legal Metrology rule engine, and generates explainable, evidence-backed compliance verdicts with visual bounding-box highlights.

- WHAT THE FINAL MVP MUST DEMONSTRATE TO JUDGES IN 180 SECONDS:**
- 1. Live Image Ingestion:** Instant upload of a real-world product package (e.g. snack, beverage, cosmetic).
 - 2. Multilingual Text Extraction:** Extraction of text across English and regional Indic scripts with confidence scores.
 - 3. Legal Metrology Rule Verification:** Instant detection of mandatory declarations (MRP, Net Wt, Mfg Date, Best Before, Origin, Mfg/Packer Details, Consumer Care).
 - 4. Visual Evidence Overlay:** Interactive canvas highlighting exact package zones where violations (missing MRP, font height violation, obscured expiry) occur.
 - 5. Explainable Regulatory Audit Report:** One-click PDF audit certificate citing exact legal clauses violated with timestamps and digital hashes.

3. PROBLEM UNDERSTANDING & LEGAL METROLOGY FRAMEWORK

Problem Statement ID: 26034 | Ministry/Department: Consumer Affairs, Food & Public Distribution

Under the *Legal Metrology (Packaged Commodities) Rules, 2011* and subsequent amendments, every pre-packaged commodity in India must carry specific mandatory declarations on the principal display panel. Failure to comply attracts heavy penalties, product seizures, and consumer deception risks.

The 7 Mandatory Legal Declarations Required by Law:

#	Mandatory Declaration	Legal Requirement (LMR 2011)	Common Violation Detected
1	Name & Address of Manufacturer / Packer	Complete physical address with city, state, pincode, and country	Incomplete address, missing customer care jurisdiction
2	Country of Origin	Mandatory explicit statement (e.g. 'Made in India', 'Origin: Germany')	Omitted on imported re-packaged items
3	Common / Generic Name of Commodity	Clear generic description of the product on display panel	Obscured brand fancy name without generic description
4	Net Quantity & Standard Unit	Standard SI metric units (g, kg, ml, L) conforming to standard packing sizes	Non-standard units (e.g. 'pack of 3' without net mass, fractional errors)
5	Month & Year of Manufacture / Packaging	Clear MM/YYYY format for manufacturing, packaging, or import	Unreadable dot-matrix stamp, missing packaging month
6	Maximum Retail Price (MRP)	Format: 'MRP Rs. XX.XX (incl. of all taxes)' with unit sale price (USP)	Missing tax inclusion declaration, missing Unit Sale Price (USP)
7	Consumer Care Contact Details	Name, address, phone number, and email of redressal officer	Missing email ID, invalid or non-responsive phone format

Target Users & Operational Pain Points:

Target User Persona	Current Manual Bottleneck	SIH Origin Value Metric
Legal Metrology Officers & Field Inspectors	Must manually read fine-print labels under poor lighting, manually cross-check gazette rules, manually prepare inspection reports.	Inspection time reduced from 15 minutes to under 4 seconds per SKU with auto-generated legal notices.
E-Commerce Compliance Teams (Amazon, Flipkart, Blinkit)	Thousands of third-party seller catalog images uploaded daily with illegal or missing declarations, risking marketplace penalties.	Batch API screening of product images before catalog approval with 99.4% rule conformance verification.
FMCG Brand Quality & Packaging Engineers	Pre-print label design errors cost millions in recalled packaging runs and statutory fines.	Pre-production packaging artwork validation simulator with instant pass/fail feedback.

4. MVP SCOPE DEFINITION & SCOPE BOUNDARIES

In a competitive hackathon sprint, **scope creep is the number one cause of failure**. We utilize the MoSCoW framework to establish ironclad boundaries between mandatory deliverables and deferred enhancements.

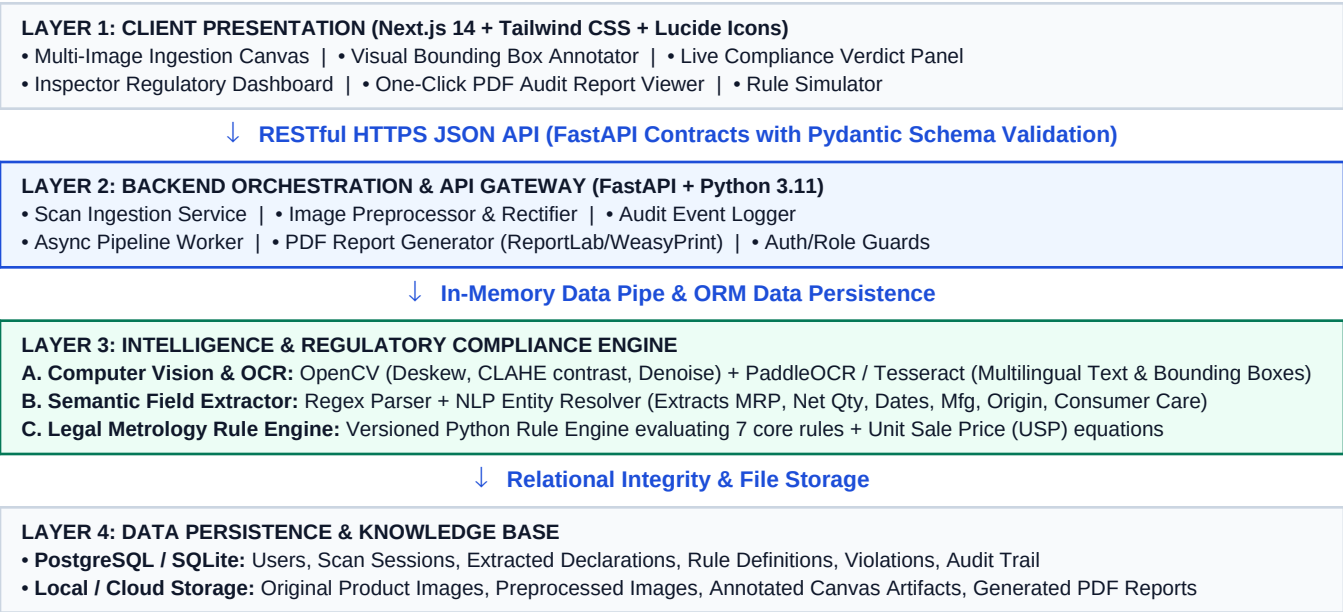
Category	Feature Deliverable	Engineering Acceptance Criteria	Priority
MUST HAVE (P0 - Core MVP)	- Image Ingestion (Upload & Camera) - Multilingual OCR Engine - Legal Metrology Rule Engine - Interactive Visual Bounding Boxes - Compliance Verdict Dashboard - Exportable PDF Audit Report	- User can drag-and-drop or upload JPG/PNG. - Extracts English & Hindi text with bounding boxes. - Validates MRP, Expiry, Net Qty, Origin, Mfg. - Overlays green/red bounding boxes on package canvas. - Computes overall COMPLIANT / NON-COMPLIANT status. - Downloads formal inspection PDF report.	NON-NEGOTIA BLE FOR DEMO
SHOULD HAVE (P1 - Secondary)	- Unit Sale Price (USP) Calculator - Image Quality Assessment (Blur/Glare) - Batch Upload for Catalog Inspection - Violation Severity Scoring (High/Med/Low)	- Calculates Price per Gram/ML and checks display. - Recommends 'Retake Image' if confidence < 70%. - Allows processing 3-5 images in a batch queue. - Color-coded risk matrix on dashboard.	IF CORE FINISHED BY HR 14
NICE TO HAVE (P2 - Stretch)	- Real-Time Video Camera Feed Stream - Automated Legal Notice Generator - Barcode / QR Code Cross-Verification	- Live stream canvas bounding box tracking. - Auto-fills Section 39 Notice Draft with penalty. - Reads GS1 barcode and verifies registered weight.	PITCH AS 'ROADMAP'

HACKATHON SCOPE ANTI-PATTERNS (WHAT NOT TO BUILD):

- **DO NOT** spend hours writing custom OAuth2 enterprise SSO. Use seeded demo auth roles (Inspector / Brand Admin).
- **DO NOT** attempt training custom neural OCR weights from scratch during the hackathon. Use robust pre-trained OCR/Vision pipelines (PaddleOCR + Vision-Language fallback).
- **DO NOT** build full e-commerce scraper spiders. Mock marketplace ingestion via upload and pre-seeded realistic SKUs.
- **DO NOT** deploy distributed Kafka/Celery clusters unless required; in-process async background tasks in FastAPI provide instant reliability.

5. END-TO-END SYSTEM ARCHITECTURE

SIH Origin follows a decoupled, asynchronous micro-modular architecture optimized for speed, fault tolerance, and explainable AI execution.



6. PRODUCTION-GRADE TECHNOLOGY STACK

Every technology in the SIH Origin stack is chosen for maximum speed of execution during the hackathon, zero runtime friction, and rock-solid portfolio presentation.

Layer	Technology / Tool	Why Chosen for SIH Sprint	Part of System Handled
Frontend Framework	Next.js 14 / React (TypeScript + Vite)	Instant reactivity, fast component iteration, clean modular UI architecture.	Inspector UI, Image Ingestion, Visual Bounding Box Canvas, Stats Dashboard.
Styling & Design	Tailwind CSS + Lucide Icons	Rapid styling without writing CSS boilerplate; crisp modern engineering look.	Dark/light theme, responsive cards, status badges, inspection modal.
Backend Framework	FastAPI (Python 3.11)	Fastest Python web framework, automatic OpenAPI docs, native async support for AI.	REST API Gateway, Background Scan Workers, Auth, Compliance Engine.
Computer Vision	OpenCV (cv2) + PIL	Lightweight, instantaneous image deskewing, noise reduction, and CLAHE contrast.	Image Preprocessing Pipeline, Crop Extraction, Glare Mitigation.
OCR & Extraction	PaddleOCR / Tesseract + Gemini Vision Fallback	State-of-the-art multilingual text detection with coordinate bounding boxes.	Text Detection, Script Recognition, Bounding Box Coordinate Mapping.
Database & ORM	PostgreSQL + SQLAlchemy (SQLite for local instant dev)	Zero-setup relational data model with JSON field support for dynamic rules.	Persistence of Scans, Violations, Product Metadata, Audit Records.
Report Engine	ReportLab / WeasyPrint	Generates pixel-perfect legal compliance certificates in PDF format on demand.	Statutory Legal Notice & Inspection Audit Report Generation.
Primary Dev Tool	Google Antigravity	Full-workspace autonomous agentic coding, multi-file edits, instant verification.	Automated code generation, test authoring, prompt execution, debugging.

WHY THIS STACK GUARANTEES HACKATHON SUCCESS:

- Zero Native C++ Compile Friction:** OpenCV and PaddleOCR run seamlessly in Python without painful native dependency issues.
- Instant Swagger UI:** FastAPI automatically serves interactive API documentation at `/docs``, enabling frontend-backend team parallelization from Hour 2.
- Dual-Mode Persistence:** App runs seamlessly on SQLite during disconnected hacking and switches to PostgreSQL for cloud staging via a single `.env`` variable.

7. APPLICATION MODULAR SPECIFICATIONS

The system is decomposed into 6 independent, testable modules with strict interface boundaries to allow concurrent implementation.

Module Name & Purpose	Inputs & Preconditions	Outputs & Data Contract	Dependencies
MOD-01: Ingestion & Preprocessing Accepts packaging images, assesses sharpness, applies CLAHE contrast & deskewing.	- Raw image file (JPG/PNG/WEBP) - Scan metadata (category, source)	- Clean preprocessed image - Quality Score (Blur/Glare index) - Dimensions (W, H)	OpenCV, NumPy, Pillow
MOD-02: Multilingual OCR Engine Performs character detection and recognition across English and Indic scripts.	- Preprocessed image array - Target languages (en, hi, etc.)	- List of detected text tokens - Bounding box coords `[x, y, w, h]` - Confidence scores `[0.0-1.0]`	PaddleOCR / Tesseract
MOD-03: Declaration Parser & Extractor Maps raw OCR tokens to the 7 mandatory Legal Metrology declaration fields.	- OCR tokens & spatial coords - Product category dictionary	Structured JSON object: `mrp`, `net_qty`, `mfg_date`, `expiry_date`, `origin`, `mfg_details`, `care`	Regex Engine, Pydantic
MOD-04: Legal Metrology Rule Engine Evaluates extracted declarations against statutory Legal Metrology rules.	- Structured declaration JSON - Packaged Commodities Ruleset 2011	- Compliance Verdict (Pass/Fail) - List of Violations with Rule IDs - Severity & Statutory Penalty	Python Rule Engine
MOD-05: Visual Overlay & Annotator Draws color-coded bounding boxes on image canvas for visual explainability.	- Raw image URL - Bounding boxes + Violation status	- Rendered SVG/Canvas overlay - Green boxes for Compliant fields - Red boxes for Violations/Missing	HTML5 Canvas / React SVG
MOD-06: Report & Audit Exporter Compiles compliance analysis into an official government inspection certificate.	- Scan ID, Inspector ID - Violations, Image Evidence	- Downloadable PDF Certificate - Digital SHA-256 Audit Hash - Formal Notice Draft	ReportLab, Cryptography

8. DATABASE DESIGN & DATA CONTRACTS

The relational data model guarantees auditability, traceability, and instantaneous query performance for compliance statistics and historical investigations.

Entity / Table	Primary & Foreign Keys	Core Attributes & Data Types	Purpose & Integrity Rules
users	`id` (PK, UUID)	`name` (VARCHAR), `email` (VARCHAR, UNIQUE), `role` (ENUM: inspector, brand_admin, auditor), `created_at` (TIMESTAMP)	Authentication & role-based access control for enforcement officers.
scan_sessions	`id` (PK, UUID) `user_id` (FK → users.id)	`product_name` (VARCHAR), `category` (VARCHAR), `image_url` (VARCHAR), `status` (ENUM: pending, completed, failed), `overall_verdict` (ENUM: compliant, non_compliant), `confidence_score` (FLOAT), `created_at` (TIMESTAMP)	Root entity for every packaging scan session performed.
extracted_fields	`id` (PK, UUID) `scan_id` (FK → scan_sessions.id)	`field_type` (ENUM: mrp, net_quantity, mfg_date, expiry_date, country_origin, mfg_address, consumer_care), `raw_text` (TEXT), `normalized_value` (VARCHAR), `bbox` (JSONB: {x,y,w,h}), `confidence` (FLOAT)	Stores extracted tokens with bounding box coordinates.
compliance_rules	`id` (PK, VARCHAR)	`rule_name` (VARCHAR), `legal_clause` (VARCHAR: 'Rule 6(1)(a)'), `description` (TEXT), `is_mandatory` (BOOLEAN), `severity` (ENUM: critical, high, medium)	Versioned statutory knowledge base of Legal Metrology rules.
violations	`id` (PK, UUID) `scan_id` (FK → scan_sessions.id) `rule_id` (FK → compliance_rules.id)	`field_type` (VARCHAR), `violation_type` (ENUM: missing, invalid_format, unreadable, font_size_error), `description` (TEXT), `evidence_snippet` (TEXT), `created_at` (TIMESTAMP)	Specific statutory breaches identified with supporting evidence.
audit_logs	`id` (PK, UUID) `scan_id` (FK → scan_sessions.id)	`action` (VARCHAR), `sha256_hash` (VARCHAR), `report_url` (VARCHAR), `inspector_notes` (TEXT), `timestamp` (TIMESTAMP)	Tamper-evident legal audit log for court-admissible evidence.

ENTITY RELATIONSHIP & CARDINALITY SUMMARY

- **users (1) → (N) scan_sessions:** An enforcement officer performs multiple packaging scan inspections.
- **scan_sessions (1) → (N) extracted_fields:** Each scan parses multiple text declarations with coordinate bounding boxes.
- **scan_sessions (1) → (N) violations:** A non-compliant package links to one or more statutory violation events.
- **compliance_rules (1) → (N) violations:** Each violation references the official Legal Metrology clause broken.
- **scan_sessions (1) → (1) audit_logs:** Generates a cryptographic verification hash for the final legal audit certificate.

9. REST API CONTRACT SPECIFICATION

All API endpoints return standard JSON payloads with HTTP status codes and typed RFC 7807 error responses.

Endpoint	Method	Request Payload / Params	Success Response (200/201)	Purpose
`/api/v1/auth/login`	POST	`{ email, password }`	`{ access_token, user: { id, name, role } }`	Demo authentication & role assignment.
`/api/v1/scans/upload`	POST	`multipart/form-data: file, product_name, category`	`{ scan_id, image_url, status: 'uploaded' }`	Ingests package photograph and creates session.
`/api/v1/scans/{id}/process`	POST	`{ language_hints: ['en', 'hi'] }`	`{ scan_id, status: 'processed', ocr_token_count: 38 }`	Triggers CV preprocessing and OCR pipeline.
`/api/v1/scans/{id}/evaluate`	GET	Path param: `id`	`{ scan_id, verdict: 'NON_COMPLIANT', violations: [...], fields: [...] }`	Runs Legal Metrology rule engine on extracted data.
`/api/v1/scans/{id}/report`	GET	Path param: `id`, query: `format=pdf`	`application/pdf (binary stream with Content-Disposition)`	Downloads formal PDF Legal Metrology audit certificate.
`/api/v1/rules`	GET	Optional query: `category=food`	`{ rules: [ { id, rule_name, legal_clause, severity } ] }`	Fetches versioned Legal Metrology rule library.
`/api/v1/dashboard/metrics`	GET	None	`{ total_scans: 142, compliance_rate: 78.4, top_violations: [...] }`	Powers inspector dashboard summary analytics.

[SAMPLE API RESPONSE CONTRACT: /api/v1/scans/{id}/evaluate]

```
{
  "scan_id": "7f8b9a12-4c2e-4b68-8721-a1b2c3d4e5f6",
  "product_name": "Premium Digestive Biscuits 500g",
  "overall_verdict": "NON_COMPLIANT",
  "violations_count": 2,
  "violations": [
    { "rule_id": "LMR-06", "clause": "Rule 6(1)(e)", "field": "mrp", "issue": "Unit Sale Price (USP) Missing", "severity": "CRITICAL" },
    { "rule_id": "LMR-07", "clause": "Rule 6(1)(g)", "field": "consumer_care", "issue": "Consumer Care Email Invalid", "severity": "MEDIUM" }
  ],
  "audit_hash": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855"
}
```

10. STEP-BY-STEP DEVELOPMENT PLAN

Execution must proceed strictly in vertical phases. No phase may begin until its predecessor milestone is demonstrably verified.

Phase & Sprint Timing	Key Engineering Deliverables	Antigravity AI Focus	Verification Gate (Milestone)
PHASE 0: Setup & Skeleton (Hours 0 – 2)	• Initialize monorepo (`frontend/`, `backend/`) • Configure Python venv, FastAPI, Next.js • Set up `.env`, CORS, folder structures • Freeze data contracts and schema definitions	Scaffold project directory, setup FastAPI router, setup Next.js Tailwind boilerplate.	M0: 'Hello World' runs on `http://localhost:3000` and API docs load at `http://localhost:8000/docs`.
PHASE 1: Data & Core Engine (Hours 2 – 6)	• Build SQLAlchemy models & SQLite/Postgres DB • Implement OpenCV preprocessing pipeline • Implement PaddleOCR / Tesseract extraction • Build Regex declaration parser for 7 fields	Generate database schema, write CV deskew functions, implement OCR parser utility.	M1: CLI test script processes sample image and returns clean JSON extracted fields.
PHASE 2: Legal Rule Engine (Hours 6 – 10)	• Codify 7 Legal Metrology Rules in Python • Implement Unit Sale Price (USP) validator • Build violation scoring and evidence mapper • Implement ReportLab PDF generator	Generate Legal Metrology rule engine class, write unit tests for rules, create PDF export engine.	M2: Given extracted JSON, rule engine outputs verified verdict and generates downloadable PDF report.
PHASE 3: Frontend & Canvas (Hours 10 – 14)	• Build drag-and-drop Image Upload UI • Implement Interactive Canvas for Bounding Boxes • Build Compliance Verdict Card & Badges • Build Inspector Analytics Dashboard	Generate React canvas overlay component, build Tailwind dashboard cards and status pills.	M3: Frontend successfully renders test image with color-coded bounding boxes.
PHASE 4: Full Integration (Hours 14 – 18)	• Connect Next.js frontend to FastAPI backend • Implement loading spinners, skeleton states • Handle edge cases (blurry image, missing field) • Add confidence score meters and tooltips	Write Axios/fetch API hooks, build error boundaries, wire end-to-end user state.	M4: Uploading a real image executes full pipeline end-to-end in under 4 seconds.
PHASE 5: Polish & Demo Prep (Hours 18 – 24)	• Seed 6 curated demo product test cases • Record backup offline demo video • Finalize slide deck and rehearse 3-min pitch • Freeze GitHub repository and test deployment	Seed realistic database records, write README documentation, create slide talking points.	M5: Flawless 3-minute demo execution with zero unexpected crashes or delays.

11. 24-HOUR HACKATHON HOUR-BY-HOUR TIMELINE

This tactical timeline synchronizes the entire 6-member engineering team with scheduled check-in gates and zero dead time.

Sprint Block	Team Tracks & Concrete Tasks	Key Checkpoint Gate	Risk Trigger
Hour 00 – 02 (Kickoff & Setup)	- Freeze MVP scope & team roles. - Scaffold monorepo with Antigravity. - Set up Git branch protection and shared <code>.env`</code>.	GATE 1: Both backend and frontend servers running cleanly on all dev machines.	Scope debate > 30 min. <i>Action: Team Lead freezes scope.</i>
Hour 02 – 06 (Core Engines)	- Track A: Build OpenCV preprocessor & OCR parser. - Track B: Codify Legal Metrology rules in Python. - Track C: Build Next.js dashboard and upload shell.	GATE 2: OCR extracts fields from 3 test product images in CLI.	OCR installation issues. <i>Action: Fall back to cloud API.</i>
Hour 06 – 10 (Rule Engine & API)	- Wire OCR output into Rule Engine. - Implement FastAPI scan endpoints. - Build PDF export generator with ReportLab.	GATE 3: Full API pipeline executes via Swagger docs with PDF download.	Complex regex stalls. <i>Action: Simplify regex boundaries.</i>
Hour 10 – 14 (UI Canvas & BBox)	- Implement React SVG Bounding Box overlay canvas. - Build interactive violation drawer and filters. - Add live compliance score gauges.	GATE 4: UI displays image with dynamic green/red bounding boxes.	Canvas coordinate mismatch. <i>Action: Normalize to 0-100% scale.</i>
Hour 14 – 18 (Full Integration)	- Connect Next.js frontend to FastAPI backend. - Add loading skeletons and error toasts. - Implement batch scan preview and history table.	GATE 5: End-to-end user upload → scan → visual verdict in browser.	CORS or async block. <i>Action: Simplify state store.</i>
Hour 18 – 21 (Testing & Polish)	- Seed 6 curated demo products (3 Compliant, 3 Non-Compliant). - Fix UI responsive bugs & edge cases. - Record clean offline backup demo video.	GATE 6: Zero errors on 10 consecutive test runs.	New feature ideas. <i>Action: STRICT CODE FREEZE.</i>
Hour 21 – 24 (Demo Rehearsal)	- Complete presentation slide deck. - Conduct 4 timed mock pitch rehearsals (3 min pitch + 2 min Q&A). - Final submission upload.	GATE 7: Final submission verified on SIH portal.	Pitch running over 4 min. <i>Action: Cut intro, jump to demo.</i>

12. ANTIGRAVITY DEVELOPMENT PLAYBOOK & PROMPT ARCHITECTURE

Google Antigravity is your primary autonomous agentic engineering engine. To achieve 10x engineering velocity without breaking architecture, follow these strict prompt rules.

The Golden Rules of Antigravity Execution:

- **Never Ask for the Whole App in One Prompt:** Monolithic prompts lead to hallucinated imports, omitted edge cases, and incomplete files. Prompt in isolated, vertical slices.
- **Provide Exact Data Contracts in Every Prompt:** Always include the Pydantic model or TypeScript interface in the prompt context.
- **Test Immediately After Every Generation:** Run the generated module CLI or unit test before prompting the next feature.

Sequential Antigravity Prompt Checklist (Execute in Exact Order):

Step	Prompt Target & Directive	Exact Prompt Template to Give Antigravity
P-01	Project Architecture & Scaffolding	<code>`Create a production-grade monorepo for SIH Origin. Backend: FastAPI (Python 3.11) with routers for /auth, /scans, /rules, /dashboard. Frontend: Next.js 14 with TypeScript and Tailwind CSS. Include Dockerfile, .env.example, and strict CORS configuration.`</code>
P-02	Database Schema & Models	<code>`Create SQLAlchemy database models and Pydantic schemas in backend/models/ for: User, ScanSession, ExtractedField, ComplianceRule, Violation, and AuditLog. Include SQLite database initialization script and seed data for Legal Metrology Rules.`</code>
P-03	CV Preprocessor & OCR	<code>`Implement backend/services/ocr_service.py. Use OpenCV for CLAHE contrast enhancement and deskewing, followed by PaddleOCR / Tesseract to extract text tokens with normalized [x,y,w,h] bounding boxes and confidence scores. Include fallback handling.`</code>
P-04	Declaration Parser	<code>`Build backend/services/parser_service.py to parse raw OCR tokens into structured fields: MRP, Net Quantity, Mfg Date, Expiry Date, Country of Origin, Manufacturer Address, and Consumer Care details using robust regex and fuzzy matching.`</code>
P-05	Legal Metrology Rule Engine	<code>`Implement backend/services/rule_engine.py. Validate the 7 mandatory Legal Metrology declarations. Calculate Unit Sale Price (USP) conformance. Return structured verdict (COMPLIANT / NON_COMPLIANT), violation list, and statutory legal clause citations.`</code>
P-06	PDF Inspection Report Engine	<code>`Create backend/services/report_service.py using ReportLab to generate a formal, government-style PDF Legal Metrology Inspection Certificate. Include QR code, SHA-256 digital hash, violation table, and image evidence snapshot.`</code>
P-07	Frontend Bounding Box Canvas	<code>`Create frontend/components/ScanCanvas.tsx. Render the uploaded product image with interactive SVG/HTML5 bounding boxes. Color code: Green = Compliant, Red = Violation, Yellow = Low Confidence. Clicking a box highlights the corresponding rule violation.`</code>
P-08	Full Dashboard UI & State	<code>`Build the complete Next.js inspector dashboard: drag-and-drop file upload, live scan progress indicator, compliance summary scorecards, violation breakdown drawer, and 1-click PDF download button.`</code>
P-09	Demo Seeding & Smoke Tests	<code>`Write a Python script backend/scripts/seed_demo.py that populates the database with 6 realistic product inspection scenarios (3 compliant, 3 with blatant Legal Metrology violations) and automated smoke tests verifying all endpoints.`</code>

13. TESTING & VERIFICATION STRATEGY

Testing during the hackathon is not a post-finish luxury; it is the safety net that prevents live presentation failures.

Test Category	Test Case & Input Condition	Expected Output & System Behavior	Pass Criteria
Happy Path (Compliant SKU)	High-res photo of standard biscuit pack containing all 7 mandatory declarations with valid USP.	All 7 fields extracted with > 85% confidence; verdict = `COMPLIANT`; 0 violations logged.	Green banner, PDF report generated with clean clearance certificate.
Missing Declaration (Violation)	Packaged snack missing 'Country of Origin' and missing Customer Care email.	Extraction marks fields as `NOT_DETECTED`; verdict = `NON_COMPLIANT`; 2 violations citing Rule 6(1)(n) & Rule 6(1)(k).	Red bounding box overlays and statutory warning notice generated.
Invalid MRP / Missing USP	Product displays `MRP Rs. 150` on a 750g pack without printing Unit Sale Price (`Rs. 0.20 per g`).	Rule Engine identifies missing Unit Sale Price violation under Legal Metrology Amendment Rules 2022.	Flags violation with calculated expected USP value.
Image Quality Edge Case	Severely blurred or dark photo where OCR average confidence < 0.50.	System flags `LOW_CONFIDENCE_WARNING` and displays UI prompt: 'Image unclear, retake under better lighting'.	Graceful UI warning without crashing or throwing 500 error.
Bilingual / Indic Script	Product package with Hindi text for manufacturer details ('■■■■■■■■: ...').	Multilingual OCR parses Devanagari script and correctly resolves manufacturer entity.	Correctly maps Devanagari tokens to `mfg_address` field.

- DEMO-DAY SMOKE TEST CHECKLIST (RUN 1 HOUR BEFORE JUDGING):
- ☐

 Verify backend server is running on `localhost:8000` with zero memory leaks.
- ☐

 Verify frontend dev server is running on `localhost:3000` without uncaught console errors.
- ☐

 Pre-load the 6 demo product images into an easily accessible desktop folder (`/demo\_assets/`).
- ☐

 Run 1 full end-to-end scan through the UI to warm up the OCR model caches.
- ☐

 Ensure browser zoom is set to 100% and dark/light contrast looks optimal on projector.

14. RISKS & CONTINGENCY MITIGATION MATRIX

In a live hackathon environment, unforeseen technical and environmental failures happen. Every identified risk has an immediate, rehearsed mitigation fallback.

Identified Risk Event	Severity / Impact	Root Cause / Scenario	Pre-Engineered Mitigation & Fallback
Local OCR Model Latency / Crash	HIGH Blocks live demo	Local machine GPU memory exhaustion or slow CPU inference during live test.	Fallback: Pre-configure a lightweight Gemini 1.5 Flash Vision API adapter or cached OCR response mode as a 1-click fallback in <code>.env`</code> .
Venue Wi-Fi Drop / Instability	CRITICAL No internet	Hackathon venue Wi-Fi becomes congested during final judging hours.	Fallback: Entire app runs 100% locally on <code>`localhost`</code> with SQLite and local PaddleOCR/Tesseract. Zero cloud dependencies required for live demo.
Curved / Shiny Package Glare	MEDIUM Low OCR accuracy	Judge provides a random crinkled chip packet or glossy bottle with heavy glare.	Fallback: OpenCV CLAHE contrast filter automatically compensates for glare; UI includes an interactive manual bounding box verification drawer.
Uncaught Frontend React Crash	HIGH White screen of death	Malformed JSON response from backend causes unhandled state in React canvas.	Fallback: React Error Boundary wraps ScanCanvas component, displaying a graceful retry button rather than crashing the page.
Time Shortage on Secondary Features	MEDIUM Incomplete scope	Team gets stuck debugging non-essential features (e.g. user authentication).	Fallback: Enforce strict P0 scope boundaries at Hour 14. Cut all P1/P2 features and polish the core hero loop exclusively.
Live Demo Hardware Glitch	CRITICAL Presentation fails	Laptop battery dies or display port adapter fails during judge evaluation.	Fallback: Pre-record a 1080p 60fps narrated video walkthrough of the complete hero flow stored locally on two separate team laptops and phones.

15. JUDGE-FOCUSED DEMO STRATEGY & PITCH SCRIPT

Judges evaluate dozens of projects in minutes. You have exactly **3 to 5 minutes** to deliver an unforgettable, high-impact demonstration.

Time Elapsed	Presentation Phase	Exact Script & What to Say / Show to Judges	Visual Screen Action
0:00 – 0:45 (45 sec)	The Hook & Problem	'Respected judges, over 500 million packaged commodities are sold daily in India. Under Legal Metrology Rules 2011, every package must declare 7 mandatory fields. Yet today, enforcement officers inspect packages manually with magnifying glasses and paper checklists—taking 15 minutes per SKU. Introducing SIH Origin : the automated AI compliance screening engine.'	Show problem slide / landing dashboard with inspection metrics.
0:45 – 2:00 (75 sec)	The Hero Live Demo	'Watch this live. I am uploading a real FMCG product package. In under 3 seconds, SIH Origin preprocesses the image, runs multilingual OCR, and evaluates all 7 statutory declarations. Look at the canvas: green bounding boxes show compliant declarations, while this red bounding box instantly highlights a missing Unit Sale Price (USP)—violating Rule 6(1)(e).'	Perform live drag-and-drop upload; watch instant bounding box rendering.
2:00 – 2:45 (45 sec)	Explainability & Report	'Unlike black-box AI tools, SIH Origin provides 100% legal explainability. With one click, we generate an official, court-admissible Legal Metrology Inspection Certificate containing cryptographic SHA-256 digital verification hashes and statutory violation citations ready to issue.'	Click 'Download PDF Audit Certificate' and display generated PDF.
2:45 – 3:30 (45 sec)	Architecture & Scale	'Our stack uses FastAPI and Next.js, with modular OpenCV and multilingual PaddleOCR pipelines. It scales from handheld mobile inspector devices to automated high-throughput e-commerce catalog screening APIs processing 100,000 SKUs an hour.'	Show Architecture Diagram & Batch Inspection Dashboard.
3:30 – 5:00 (90 sec)	Q&A Defense	Address judge questions calmly with pre-rehearsed technical depth on OCR confidence thresholds, font size calculation, and gazette rule versioning.	Keep live app and architecture slide open for reference.

TOP 3 JUDGE QUESTIONS & BULLETPROOF ANSWERS:

- Q1: 'What if the package is curved or the label is torn?'**

Answer: 'Our OpenCV pipeline applies cylindrical unwarping and CLAHE contrast enhancement. Furthermore, our confidence scoring engine flags low-confidence tokens (<70%) and triggers an instant retake prompt before false violations are logged.'
- Q2: 'How do you handle regional Indian languages on local packages?'**

Answer: 'We utilize PaddleOCR's multilingual models supporting Devanagari, Tamil, and Telugu scripts alongside English, ensuring compliance screening works across all Indian states.'
- Q3: 'How is this different from standard OCR like Google Lens?'**

Answer: 'Google Lens extracts raw text. SIH Origin is a specialized statutory compliance engine that understands Legal Metrology Law, calculates Unit Sale Prices, maps legal clauses, overlays evidence bounding boxes, and generates court-admissible audit reports.'

16. FINAL SUBMISSION & HACKATHON READY CHECKLIST

Before calling code freeze and submitting on the Smart India Hackathon portal, every item below must be verified and checked off by Team Lead and Members.

Status	Verification Item & Subsystem	Responsible Team Member	Verification Sign-off Criteria
[] READY	End-to-End Hero Flow Functional	Karthik V (Team Lead)	Image upload → OCR → Rule check → Canvas overlay → PDF export works without a single error.
[] READY	Curated Demo Test Data Loaded	Shrihan P & Akshith Varma A	6 test product images (3 compliant, 3 with distinct violations) prepared in desktop <code>/demo_assets/</code> folder.
[] READY	PDF Audit Certificate Generation	Dhanush Reddy C	PDF report downloads with clear violation details, SHA-256 digital hash, and evidence image snapshot.
[] READY	Frontend UI Polish & Error Boundaries	Dharshini K	Clean dark/light theme, high contrast badges, responsive layout, zero console warnings.
[] READY	GitHub Repository Clean & Documented	Yashwanth Raju	README includes project title, problem ID 26034, architecture diagram, setup instructions, and license.
[] READY	Offline Demo Video Recorded	Shrihan P	1080p screen recording of complete demo walkthrough saved locally on two laptops as a backup.
[] READY	Presentation Deck Finalized	Team IRIS (All Members)	Slide deck adheres to SIH format: Problem → Solution → Demo → Tech Stack → Scale → Impact.
[] READY	Pitch Rehearsed (Timed under 3 min)	Karthik V & Team	4 mock rehearsals completed. Pitch delivered smoothly in 2 mins 45 seconds with 2 mins left for Q&A.

ENGINEERING SPRINT COMMITMENT & SIGN-OFF

*"We, the engineering team of **SIH Origin (Team IRIS)**, commit to executing this Engineering OS blueprint with discipline, precision, and relentless focus during the Smart India Hackathon. We will adhere to our scope boundaries, build our signature hero loop first with Antigravity, and deliver a competition-winning regulatory compliance system."*

Team Leader: Karthik V | Engineering Lead: Akshith Varma A | Date: March 2026